TRACE: Commercialization Plan

Executive Summary

TRACE (Triage, Resolution, Audit, and Compliance Engine) is an AI-powered field service platform that guides junior diesel technicians through fault code diagnosis with built-in human oversight. By combining local open-source LLMs with a structured multi-agent workflow, TRACE reduces Mean Time to Resolve by 33% while ensuring every safety-critical repair is approved by a named senior engineer. The platform is designed for Cummins-authorized dealers, fleet operators, and OEM service networks.

Target Market

Primary: Cummins-Authorized Dealers

There are approximately 3,500 Cummins-authorized service locations in North America. Each employs 5 to 30 field technicians. Junior techs (under 2 years experience) make up roughly 40% of the workforce and generate the most escalation calls to senior engineers. TRACE directly addresses this bottleneck.

Secondary: National Fleet Operators

Large fleets (Ryder, Penske, Werner, Schneider) maintain thousands of Cummins-powered vehicles in-house. These operators face the same junior-tech-to-senior-engineer bottleneck at scale, with the added pressure of minimizing vehicle downtime across distributed locations.

Tertiary: Other OEM Service Networks

The TRACE architecture is engine-agnostic. The fault code database and prompt templates can be adapted for Caterpillar, Detroit Diesel, Volvo, and other OEM service networks. Each OEM represents a separate addressable market of similar size.

Competitive Landscape

	ServiceMax	**Augmentir**	**TRACE**
AI Diagnosis	No	Basic (generic)	Yes (fault-code-specific LLM)
Human-in-the-Loop	No	No	Yes (mandatory for safety)
Offline Support	Limited	No	Full (local LLM via Ollama)
Audit Trail	Basic logging	Basic logging	Append-only, per-agent decision log
LLM Cost	N/A	Cloud API fees	Zero (open-source, local)
Deployment	Cloud-only SaaS	Cloud-only SaaS	On-premise or hybrid

Key Differentiators

1. **Zero per-query AI cost.** TRACE uses open-source LLMs (Llama 3.1, Mistral 7B) running locally. Competitors using cloud APIs (GPT-4, Claude) incur \$0.01 to \$0.10 per query. At 50 techs making 10 queries/day, this saves \$12,000 to \$125,000 per year.
2. **Offline-first architecture.** Field techs frequently work in remote areas with no connectivity. TRACE runs entirely on-device when offline and syncs when reconnected. No competitor offers full AI diagnosis without internet.
3. **Mandatory human oversight for safety.** TRACE is the only platform that programmatically enforces human approval for high-risk, high-cost, or low-confidence repairs. This is critical for OEM warranty compliance and liability protection.

Go-to-Market Strategy

Phase 1: Direct OEM Partnership (Year 1)

Approach Cummins Inc. directly through their distributor network. Offer TRACE as a white-label solution embedded in Cummins' existing service tools (QuickServe Online, INLINE diagnostics). Revenue model: licensing fee per dealer location.

Why Cummins first: TRACE is already built around Cummins fault codes (P0191, ISB engine). The demo is Cummins-specific. This creates a natural entry point and reference customer.

Phase 2: Fleet Operator Sales (Year 2)

Target the top 20 national fleet operators who maintain Cummins-powered vehicles in-house. These operators have centralized procurement and can deploy TRACE across hundreds of locations simultaneously. Revenue model: per-seat subscription.

Phase 3: Multi-OEM Expansion (Year 3)

Adapt TRACE for Caterpillar, Detroit Diesel, and Volvo fault code systems. Each OEM adaptation requires a new fault code database and prompt template set (approximately 4 to 6 weeks of development per OEM). Sell through OEM distributor partnerships.

Value Proposition

TRACE solves a quantifiable problem: junior diesel technicians (under 2 years experience) account for 40% of the workforce but generate over 70% of escalation calls to senior engineers. Each escalation costs 30 to 90 minutes of senior engineer time, creating a bottleneck that directly increases Mean Time to Resolve (MTTR) and vehicle downtime.

For Cummins-Authorized Dealers

1. **33% reduction in MTTR.** AI-guided triage eliminates the diagnostic search process for common fault codes. Junior techs resolve more tickets independently.
2. **60% fewer escalation calls.** The multi-agent pipeline handles evidence collection, confidence scoring, and routine approvals automatically. Senior engineers only review genuinely complex or safety-critical cases.
3. **Complete audit trail for warranty compliance.** Every diagnostic decision is logged with the agent that made it, the evidence used, and the human who approved it. This satisfies OEM warranty audit requirements without additional paperwork.

For National Fleet Operators

1. **Reduced vehicle downtime.** Faster diagnosis means trucks return to service sooner. For a fleet of 1,000 vehicles, even a 1-hour reduction in average downtime per incident translates to significant revenue recovery.
2. **Standardized diagnostic quality.** TRACE ensures every technician follows the same evidence-based diagnostic process, regardless of experience level or location. This eliminates the quality variance across distributed service centers.
3. **Zero AI infrastructure cost.** The LLM runs on the technician's existing hardware. No cloud GPU instances, no per-query API fees, no vendor lock-in.